TECHNICAL DATA ANNEX

Appendix A – Structured Long-Term Ecological Records

Huaravito Reserve LTEMS

Long-Term Ecological Monitoring Site

Piedades Sur, Costa Rica | 2021–2026

Compiled by:

Randall Santamaría Artavia

Custodian-Director

February 2026

Supporting documentation for: Biogenic Thermal Buffering Research | Hauya elegans Population Analysis | Sentinel Protocol

Table of Contents

1. Executive Data Summary

2. Dataset 1: Biodiversity Master Record (2021–2026)

3. Dataset 2: Thermal Monitoring Time Series (2022–2025)

4. Dataset 3: Thermal Gradient Transect (February 2026)

5. Dataset 4: Forest Structure & Hauya elegans Density

6. Conceptual Diagrams: Biogenic Cooling Mechanisms

7. Data Quality Statement & Limitations

8. References to Primary Reports

1. Executive Data Summary

> *This annex consolidates five years of structured field observations and thermal monitoring within the Huaravito Reserve Long-Term Ecological Monitoring Site (LTEMS). The data indicate persistent microclimatic buffering within the Toyopán Sacred Forest core, with recurrent temperature differentials relative to adjacent exposed areas. The structural dominance of Hauya elegans and the presence of a thick organic soil horizon support the hypothesis of a biogenic cooling mechanism associated with mature forest architecture.*

1.1 Summary Statistics (2021–2026)

Parameter	Value
Total iNaturalist observations	238 (as of February 2026)
Total species documented	116
Hauya elegans observations	72 (30.3% of total observations)
Thermal monitoring period	2022–2025 (4 years continuous)
Mean annual temperature (forest core)	19.5°C (σ ~0.2°C interannual)
Biogenic cooling effect (vs. matrix)	-3.7°C (sustained)
Maximum thermal gradient documented	10.2°C (Tajo to Shrine, Feb 2026)
Extreme event buffering capacity	+6.0°C (Feb 2026 polar intrusion)

2. Dataset 1: Biodiversity Master Record (2021–2026)

2.1 Annual Observation Summary

The following table summarizes iNaturalist observations by year, highlighting the increase in monitoring effort and the consistent presence of Hauya elegans across all years.

Year	Total Obs.	Species Richness	H. elegans Records	% Dominance	Notes
2021	15	12	3	20.0%	Baseline
2022	28	22	7	25.0%	—
2023	42	35	12	28.6%	—
2024	71	58	22	31.0%	—
2025	82	68	28	34.1%	—
2026 (Feb)	**238**	**116**	**72**	**30.3%**	Current

Table 2.1. Annual biodiversity monitoring summary. % Dominance = (H. elegans / Total Obs.) × 100. The increase in total observations from 2021 to 2026 reflects intensified field effort and improved taxonomic verification via iNaturalist community. The consistent ~25–34% representation of H. elegans across years suggests structural prominence within the forest core.

2.2 Interpretation

The progressive increase in total observations (15 → 238) and species richness (12 → 116) over five years reflects: (a) expanded spatial coverage of monitoring; (b) improved observer expertise in field identification; and (c) phenological variation (different seasons capture different taxa). The sustained representation of Hauya elegans at 25–34% of total observations is consistent with its role as a structurally prominent canopy species, though this does not equate to absolute numerical dominance (see Section 5 for density analysis).

3. Dataset 2: Thermal Monitoring Time Series (2022–2025)

3.1 Annual Mean Temperature Summary

Continuous temperature monitoring within the forest core (The Shrine) using a fixed-position sensor at 1.5 m height.

Parameter	2022	2023	2024	2025	Mean
Dry Season (Dec–Apr)	19.9°C	20.1°C	19.8°C	20.0°C	**20.0°C**
Wet Season (May–Nov)	19.2°C	19.4°C	19.1°C	19.3°C	**19.3°C**
Annual Mean	**19.4°C**	**19.6°C**	**19.5°C**	**19.5°C**	**19.5°C**

Table 3.1. Multi-year thermal matrix. Interannual variance σ ~0.2°C. Annual thermal amplitude: 2.1°C (18.8°C min to 20.9°C max). Dry season consistently ~0.7°C warmer than wet season.

3.2 Key Thermal Events

Event	Description
Baseline stability	Mean 19.5°C maintained across 4 years with σ ~0.2°C, characteristic of intact primary cloud forest thermal homeostasis.
Biogenic cooling effect	-3.7°C relative to San Ramón regional ambient temperature (sustained differential 2022–2025).
Feb 2026 polar event	Regional ambient 12.2°C, external wind-chill ~10°C, forest core maintained 16.2°C (buffering capacity: +6°C).

4. Dataset 3: Thermal Gradient Transect (February 20, 2026)

4.1 Spatial Temperature Profile

Single-day transect documenting temperature gradient from degraded land (abandoned clay extraction site) to mature forest interior. Measurements taken midday under partly cloudy conditions with light north wind.

ID	Sampling Point	Temp (°C)	ΔT from Tajo	Vegetation Structure
1	Abandoned Tajo (center)	**27.5**	0.0	Bare clay, full sun
2	Tajo perimeter	26.4	-1.1	Grassland buffer
3	East entrance	23.6	-3.9	Young secondary (~1.5 m shrubs)
4	West entrance	23.5	-4.0	Transition to mature canopy
5	Mature Hauya elegans tree	21.6	-5.9	Old-growth with buttress roots
6	Reserve deep core	20.0	-7.5	High biomass, closed canopy
7	The Shrine (forest interior)	**17.3**	**-10.2**	Max canopy closure, 20 cm organic layer

Table 4.1. Thermal gradient transect (~150 m horizontal distance). Total gradient: 10.2°C. Note: Single-day observation; temporal replication required to confirm reproducibility (see limitations, Section 7).

4.2 Conceptual Cross-Section

The temperature gradient follows a progressive increase in vegetation structural complexity:

- Degraded zone (27.5°C): Bare soil, high albedo, full solar radiation exposure
- Edge transition (23–26°C): Partial canopy cover, reduced wind velocity
- Mature forest core (17–22°C): Closed canopy, high epiphytic biomass, deep organic soil, evapotranspirative cooling

This spatial gradient provides a "natural experiment" for quantifying the biogenic cooling effect of mature forest structure.

5. Dataset 4: Forest Structure & Hauya elegans Density

5.1 Spatial Distribution of Hauya elegans

The 72 documented H. elegans observations are not uniformly distributed across the 5,800 m² reserve but are concentrated in the western sector (The Shrine / Toyopán core), where canopy closure and organic soil depth are greatest.

Zone	Area (m²)	H. elegans obs.	Density (obs./ha)
Western Core (The Shrine)	~2,000	58	**290**
Transition Zone	~1,800	12	67
Edge / Regeneration Zone	~2,000	2	10
Total Reserve	**5,800**	**72**	**124**

Table 5.1. Spatial distribution of H. elegans observations. Note: These are observed/documented individuals (likely large, conspicuous trees), not complete census. Actual density may be higher if small individuals are underrepresented. The 290 obs./ha in the Western Core is exceptionally high for a regionally rare species.

5.2 Interpretation: Old-Growth Hypothesis

The high concentration of H. elegans in the Western Core, combined with structural attributes (buttressed roots, >70% epiphytic cover, 20 cm organic soil layer, absence of compaction), is consistent with the hypothesis that this zone represents a minimally disturbed old-growth remnant rather than recovering secondary forest. However, definitive confirmation requires dendrochronological analysis (tree ring dating).

6. Conceptual Diagrams: Biogenic Cooling Mechanisms

6.1 Proposed Mechanisms (Integrated System)

The following conceptual framework synthesizes field observations into a testable hypothesis for the observed thermal stability (σ ~0.2°C) and cooling effect (-3.7 to -10.2°C).

MECHANISM 1: Orographic Fog Interception (NE Boundary)

Topographic channeling of cool, moist air from abandoned mine (north) toward NE forest boundary. Hauya elegans canopy + epiphytic biomass intercepts fog, condensing water vapor and releasing latent heat. Creates localized evaporative cooling persisting into dry season.

MECHANISM 2: Epiphytic-Bryophytic Thermal Buffer

>70% trunk surface coverage by bryophytes, orchids, ferns increases thermal inertia of tree trunks. High water content of saturated mosses buffers against rapid temperature fluctuations. Secondary shade layer (vines, lianas) reduces wind penetration, trapping cool air mass in understory.

MECHANISM 3: Convective Circulation (E-W Gradient, speculative)

Dense, cool, fog-laden air enters from NE, displaces warmer air laterally toward west. Tall canopy (west) functions as thermal chimney: warm air rises and evacuates at canopy height while understory remains cool. Continuous circulation prevents heat accumulation. NOTE: Requires distributed sensor network for validation.

MECHANISM 4: Soil-Atmosphere Interface ("Living Crust")

Dense 20 cm organic layer (roots, litter, fungal hyphae) retains soil moisture via capillary action. Continuous evaporative cooling from soil surface. Physical + allelopathic suppression of Pennisetum purpureum (<1% grass cover) prevents heat gain from thermophilic invasives.

6.2 Synergistic Integration

The 10.2°C thermal gradient and sustained -3.7°C cooling effect are hypothesized to result from the synergistic interaction of all four mechanisms operating simultaneously. No single mechanism alone would produce the observed magnitude of thermal decoupling. The system functions as an integrated biogenic heat exchanger.

7. Data Quality Statement & Limitations

7.1 Biodiversity Dataset

- Opportunistic sampling: Not systematic census. Likely underrepresents small/cryptic species.
- Detectability bias: Large, conspicuous species (H. elegans) overrepresented relative to actual abundance.
- Temporal bias: Observations concentrated in accessible periods (dry season); phenological variation may be undersampled.

7.2 Thermal Monitoring Dataset

- Single sensor: No spatial replication. Cannot confirm horizontal or vertical temperature gradients.
- Unknown instrument precision: Calibration status not documented.
- Fixed height (1.5 m): Does not capture canopy or soil surface temperatures.
- Thermal gradient (Feb 2026): Single-day observation. Reproducibility across seasons/weather conditions unknown.

7.3 Forest Structure Dataset

- No formal dendrometry: DBH not measured. Size estimates based on visual assessment.
- Incomplete spatial coverage: Small individuals and edge zones likely undersampled.
- No recruitment/mortality data: Population dynamics unknown (requires permanent plots).

7.4 Recommendations for Future Data Collection

- Systematic census with complete spatial coverage and size-class distribution
- Distributed temperature/humidity sensor network (minimum 6 nodes)
- Anemometer array for wind velocity profiling
- Permanent monitoring plots for demographic tracking
- Dendrochronological sampling (increment cores) for age structure

8. References to Primary Reports

This Technical Data Annex supports and is supported by the following primary reports prepared for Huaravito Reserve LTEMS:

1. Biogenic Thermal Buffering Along a Forest Maturity Gradient at Huaravito Reserve: Preliminary Field Observation Report (February 2026)

Documents the 10.2°C thermal gradient and proposes testable mechanisms for extreme cooling effect.

2. Proposed Mechanisms of Biogenic Cooling at Huaravito Reserve: A Five-Year Field Observation-Based Hypothesis (February 2026)

Presents the integrated hypothesis for thermal homeostasis based on orographic fog interception, epiphytic buffering, convective circulation, and soil-atmosphere interface dynamics.

3. Hauya elegans (Guayabón) at Huaravito Reserve LTEMS: Population Structure and Potential Role as an Old-Growth Indicator Species (February 2026)

Analyzes the 72 documented H. elegans observations and discusses evidence for old-growth forest status in the Toyopán Sacred Forest core.

4. Five-Year Chronosequence of Unassisted Natural Regeneration in a Premontane Forest Fragment, Costa Rica (February 2026)

Provides floristic baseline, recruitment dynamics, and microclimatic stability data submitted to SNAPP-MORE Working Group.

Compiled by: Randall Santamaría Artavia

Custodian-Director, Huaravito Reserve LTEMS

Piedades Sur, San Ramón, Costa Rica

February 2026